What is Docker?

Docker is a **containerization platform** used to package applications with all dependencies and run them anywhere.

📌 2. Difference: Docker vs Virtual Machine

Feature	Docker	Virtual Machine
Size	Small	Large
Boot time	Seconds	Minutes
OS	Shares host OS	Full OS
Performance	Fast	Slower

📌 3. What is a Container?

- Lightweight environment
- Runs application + dependencies
  👉 Example:

```
docker run nginx
```

📌 4. What is a Docker Image?

- Read-only template
- Used to create containers

👉 Example:

```
docker pull ubuntu
```

📌 5. Dockerfile

Used to build custom image

```
FROM ubuntu
RUN apt update
CMD ["echo", "Hello"]
```

👉 Build:

```
docker build -t myimage .
```

📌 6. Important Commands

```
docker ps              # running containers
docker ps -a           # all containers
docker images          # list images
docker run -it ubuntu  # run container
docker stop <id>       # stop
docker rm <id>         # remove container
docker rmi <id>        # remove image
```

📌 7. What is Docker Hub?

- Public registry to store images
  👉 Commands:

```
docker login
docker push username/image
```

📌 8. Port Mapping

Connect container to outside world

```
docker run -p 8080:80 nginx
```

📌 9. Volume (Data Persistence)

Stores data permanently

```
docker run -v /data:/app ubuntu
```

📌 10. What is Docker Compose?

- Tool to run **multi-container applications**

```
version: "3"
services:
```

```
  web:
    image: nginx
```

👉 Run:

```
docker-compose up
```

📌 11. What is Docker Swarm?

- Docker's native clustering tool
- Used for scaling containers

📌 12. Common Interview Questions

❓ What is the difference between CMD and ENTRYPOINT?

- CMD → default command (can override)
- ENTRYPOINT → fixed command

❓ What happens when you run `docker run`?

1. Pull image (if not available)
2. Create container
3. Start container

❓ Difference between Image & Container?

- Image → blueprint
- Container → running instance

❓ What is layering in Docker?

- Each instruction in Dockerfile creates a layer
- Helps caching & faster builds

❓ What is COPY vs ADD?

- COPY → simple copy
- ADD → copy + extra features (URL, extract)

📌 13. Common Errors

❌ Permission denied:

```
sudo usermod -aG docker $USER
```

📌 14. Real-Time Use Case

- Build app → create image → push to Docker Hub
- Pull image → run container in server

📌 15. One-Line Answer (Very Important)

👉 Docker is a tool that **packages applications into containers for consistent deployment across environments**